MEMORANDUM

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Date: February 16, 2026

RE: **Obtaining performance curves for fan at two operating speeds and using scaling laws to compare experimental data to predicted performance parameters**

In this experiment, performance curves for an axial fan operating at two different rotational speeds (1320 RPM and 1750 RPM) were developed using calculated values of flowrate, actual head gain, motor power, and fan efficiency. By incrementally closing the variable air volume (VAV) dampers, the mass flowrate throughout the connected piping system was adjusted, and pressure readings from Pitot-static tubes were recorded to obtain several data points used to graphically represent fan characteristics. Additionally, performance data collected at 1320 RPM were applied to scaling equations to predict the system behavior at 1750 RPM. The resulting performance curves showed that head gain decreased with increasing flowrate, while motor power increased with increasing flowrate at both operating speeds. The highest efficiency occurred at midrange flowrates, corresponding to the best efficiency point (BEP) of the fan. For the 1320 RPM condition, peak efficiency was 17.9%, and for the 1750 RPM condition, peak efficiency was approximately 18.3%, indicating that efficiency remained relatively constant with changes in rotational speed. Comparison of the scaled 1750 RPM predictions with the measured 1750 RPM data demonstrated good agreement in the midrange operating region, confirming the validity of the fan similarity laws. Deviations between predicted and measured results were most noticeable near stall conditions, where flow separation and increased losses reduce dynamic similarity.

Introduction

Axial fans are ubiquitous throughout fluids systems utilized in ventilation, cooling, and air circulation in various industries due to their ability to move large volumes of air efficiently. To maximize an axial fan's efficiency with its corresponding piping system, its operating point should be selected at or near its best efficiency point (BEP), which can be identified using performance curves. These performance curves can be built from experimental or calculated data for flowrate, motor power, and fan efficiency

Scaling laws are very helpful tools that predict a particular system's behavior based on a similar but smaller (or larger) system with known behavior. Using dimensional analysis, one can obtain dimensionless coefficients (for example, this experimental analysis uses C_H, C_Q, and $C_{\mathcal{P}}$) that relate groups of variables using the Buckingham Pi theorem.

Background

Knowing that the flowrate of the fan is split between two pipe branches (reference Figure 1), conservation of mass for an incompressible fluid can be applied to the two branches (denoted by the two corresponding Pitot-static tube locations, 8 and 9) to obtain

$$Q_{fan} = Q_8 + Q_9 \tag{1}$$

By substituting in the definition for average velocity $\bar{V}$ and assuming fully developed turbulent flow due to the high flowrates and long straight pipe sections characterizing the experiment, Equation 1 becomes

$$\frac{\bar{V}}{V_c} = \frac{2n^2}{(n+1)(2n+1)}, \quad n = -1.7 + 1.8\log\left(Re_{V_c}\right) \tag{2}$$

where the Reynolds number is defined as

$$Re_{V_c} = \frac{\rho V_c D_{duct}}{\mu} \tag{3}$$

Equations to calculate actual head rise h_a from pressure, average velocity, and known constants can be calculated from the static pressure rise across the fan and the calculated flow rate

$$h_a = \frac{\Delta p_{fan}}{\rho} + (1 + K_L)\left(\frac{\bar{V}_{fan}{}^2}{2g}\right), \tag{4}$$

assuming constant elevation ($\Delta z = 0$), ambient atmospheric air at the inlet ($p_1 = p_{atm}, \bar{V}_1 \approx 0$), and low pressure and high velocity air at the outlet ($\alpha_2 \approx 1$). Dimensional analysis and the Buckingham Pi theorem can be used to determine the dependence of variables of interest on other fan parameters to derive scaling relationships between fans.

$$h_a = f_1(Q_{fan}, \omega, D_{fan}, \rho)$$
$$\dot{W}_{shaft} = f_2(Q_{fan}, \omega, D_{fan}, \rho)$$
$$\eta = f_3(Q_{fan}, \omega, D_{fan}, \rho)$$

Procedure/Analysis

The experiment was conducted using the axial fan test stand shown in Figure 1. Prior to lab, the system was prepared by the instructor to ensure proper pressurized air supply to the variable air volume (VAV) dampers and correct configuration of the variable frequency drive (VFD). The fan was operated at 1320 RPM and 1750 RPM, controlled through the VFD. For each speed, the fan was allowed to reach steady-state operation before any measurements were recorded.

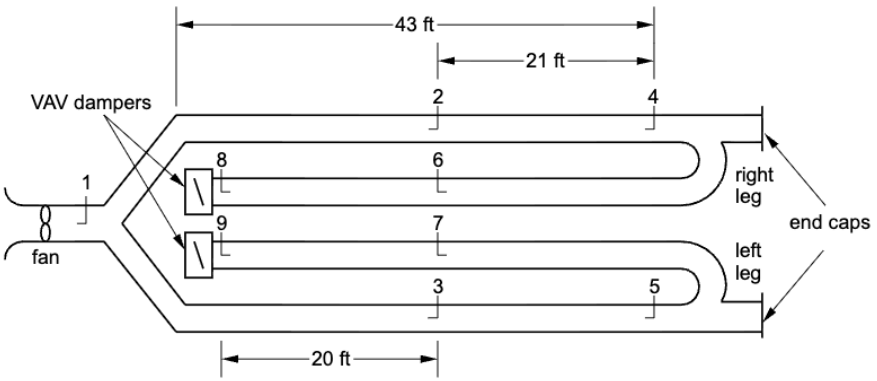


Figure 1. Axial fan test stand

At each speed, the volumetric flow rate through the fan was varied by incrementally adjusting the left VAV damper, changing the system resistance. About fifteen data recordings were recorded. At each damper setting, measurements of fan static pressure, motor voltage, motor current, and centerline velocities in both outlet ducts were recorded using the Pitot-static tubes and differential pressure gauges. The flow rate reduced until fan stall was observed, which was concluded by the change in noise and unsteady pressure readings. This process was repeated to obtain sufficient date points over a range of flow rates. The first two groups conducted these measurements at the first speed while our group along with another group conducted this same procedure at 1750 RPM. Once we obtained all our data, the fan was powered off and shut down properly.

The measurements for static pressure, left flow, and right flow were measured in inches of H_2O, so these measurements are converted to feet of air using Equation 5.

$$h_{a,air} = \left(\frac{\rho_{H_2O}}{\rho_{air}}\right) h_{a,H_2O} \tag{5}$$

The center-line velocity in the duct was calculated using the relationship between the dynamic pressure measured by the pitot-static tube and the velocity in the duct. This relationship is shown in Equation 6.

$$V_C = \sqrt{\frac{2(p_T - p)}{\rho_{air}}} \tag{6}$$

Then, Equation 2 is used to find the average velocity through the duct. The actual head rise is computed using Equation 4. Finally, the volumetric flowrate is calculated using Equation 1 after computing the individual volumetric flowrates in the two ducts.

Data was collected at two different fan speeds, but scaling laws can be used to predict the parameters of the fan at a different speed such as head gain, motor power, and efficiency. For the fans to be scaled, the dimensionless head, power, and efficiencies must match. Thus, these parameters for each pump are set equal and solved for the variable of interest. The equations for the dimensionless head, dimensionless power, and efficiency are shown below. Fan efficiency can be calculated using Equation 7,

$$\eta = \frac{\mathcal{P}_f}{\dot{W}_{shaft}} = \frac{\rho g Q_{fan} h_a}{\sqrt{3} V_{LL} I_L \cos\theta} \tag{7}$$

assuming $\dot{W}_{shaft} = \dot{W}_{in}$. The dimensionless head rise coefficient can be determined using Equation 8, and the dimensionless flowrate coefficient can be determined using Equation 9.

$$C_H = \frac{g h_a}{\omega^2 D_{fan}^2} \tag{8}$$

$$C_Q = \frac{Q_{fan}}{\omega D_{fan}^3} \tag{9}$$

Results & Discussion

The data used for the experiment analysis is shown in Attachment A. It is important to note that the data provided was used for our analysis due to the damaged experimental setup. The results for the volumetric flowrate, actual head gain, motor power, and efficiency for the axial fan at 1320 RPM are shown in Table 1. A sample calculation for these values for number 1 for the fan at 1320 RPM is shown in Attachment B.

Table 1. Volumetric Flowrate, Actual Head Gain, Motor Power, and Efficiency for Axial Fan at 1320 RPM

Number	Volumetric Flowrate, Q_{fan} (cfm)	Actual Head Gain, h_a (in H_2O)	Motor Power, $\dot{W}_{shaft}$ (hp)	Efficiency (%)
1	3312	0.261	0.909	15.0
2	3234	0.275	0.922	15.2
3	3219	0.295	0.928	16.1
4	3134	0.309	0.949	16.1
5	2957	0.317	0.971	15.2
6	2910	0.396	1.018	17.9
7	2861	0.414	1.049	17.8
8	2541	0.434	1.067	16.3
9	2404	0.467	1.058	16.7
10	2156	0.433	0.996	14.8
11	1937	0.401	0.939	13.0
12	1690	0.390	0.863	12.0
13	1552	0.363	0.828	10.7
14	1233	0.394	0.873	8.8
15	1140	0.433	0.925	8.4
16	928	0.448	0.907	7.2

As shown by the results, the actual head gain increases as the volumetric flowrate decreases. At higher flowrates, there is more friction and increased turbulence, lowering the actual head gain. Additionally, as the flow is more restricted, representing the lower flowrates, the fan will build up more pressure. The motor power is highest for moderate flowrates. Similarly, the highest efficiency is when the flowrate is more moderate. Specifically, the efficiency is highest, 17.9%, when the flowrate is equal to 2910 cfm. Operation near the BEP is preferred because it maximizes useful energy transfer while minimizing losses and mechanical stress.

The results for the volumetric flowrate, actual head gain, motor power, and efficiency for the axial fan at 1750 RPM are shown in Table 2.

Table 2. Volumetric Flowrate, Actual Head Gain, Motor Power, and Efficiency for Axial Fan at 1750 RPM

Number	Volumetric Flowrate, Q_{fan} (cfm)	Actual Head Gain, h_a (in H_2O)	Motor Power, $\dot{W}_{shaft}$ (hp)	Efficiency (%)
1	4690	0.522	2.114	18.3
2	4343	0.504	2.184	15.8
3	4292	0.603	2.256	18.1
4	3937	0.629	2.359	16.5
5	3898	0.646	2.377	16.7
6	3616	0.683	2.420	16.1
7	3496	0.776	2.494	17.1
8	3165	0.832	2.479	16.7
9	2449	0.720	2.058	13.5
10	2156	0.683	2.019	11.5
11	1908	0.649	1.937	10.1
12	1240	0.769	2.103	7.1
13	662	0.964	2.323	4.3
14	0	1.250	2.496	0.0

Similar trends for the calculated parameters that were observed when the fan was running at 1320 RPM are observed for the fan at 1750 RPM. Specifically, the head increases with decreasing flowrate. However, the efficiency at this speed was highest for the largest flowrate, 4690 cfm, indicating that this is the best efficiency point for the pump at this speed.

Figure 2 shows the actual head gain (in. H_2O) as a function of volumetric flowrate (cfm) for both 75% (1320 rpm) and 100% (1750 rpm) rotational speeds. At both speeds, the head gain decreases as flowrate increases, which is consistent with typical axial fan behavior. The higher rotational speed produces a greater head rise at all flowrates. Stall occurs when flow separation develops on the blades, resulting in unsteady pressure readings and reduced performance, more visible at lower flowrates.

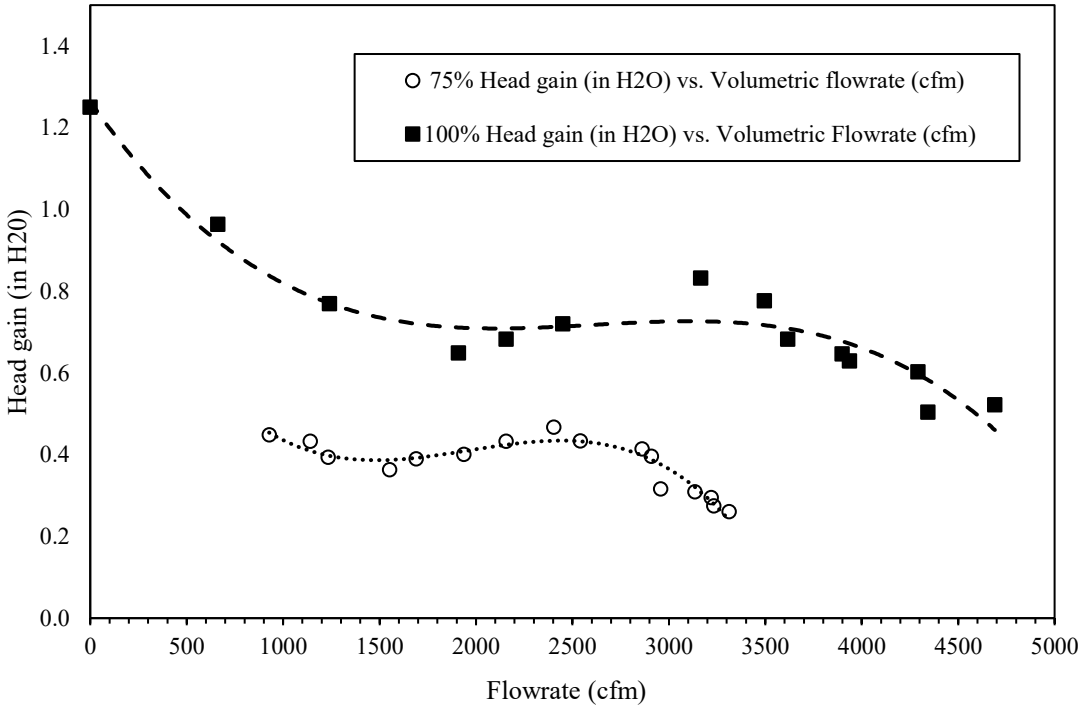


Figure 2. Head gain vs Flowrate fit with a third-order polynomial curve

Figure 3 shows motor power (hp) versus flowrate (cfm) for both rotational speeds. Motor power generally increases with increasing flowrate for each speed while also the 100% speed condition requires significantly more power than the 75% condition across all flowrates. The trendlines show a nonlinear relationship between power and flowrate, which agrees with expected fan performance characteristics because power scales strongly with rotational speed. The higher speed condition results in a substantial increase in power consumption.

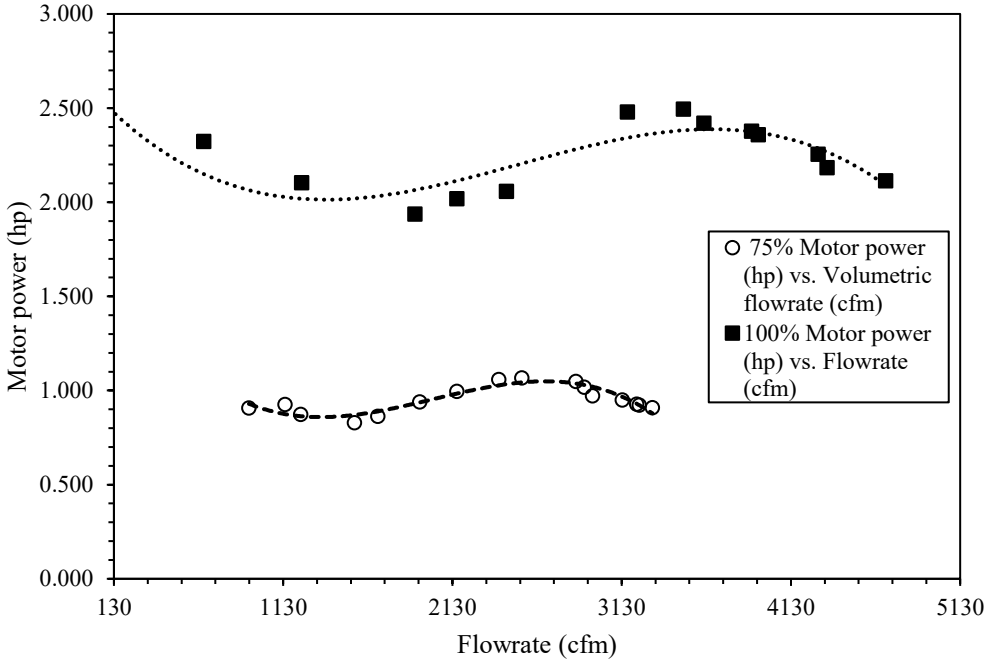


Figure 3. Motor Power vs Flowrate fit with a third-order polynomial curve

Finally, Figure 4 shows overall fan efficiency as a function of flowrate for both speeds. For each rotational speed, efficiency increases with flowrate, reaches a maximum best efficiency point (BEP), and then begins to decrease slightly. The peak efficiency occurs at an intermediate flowrate rather than at very low or very high flowrates. Operation near this peak would be preferred because it delivers the most useful energy to the fluid relative to the input power. The 100% speed condition achieves slightly higher maximum efficiency compared to the 75% condition.

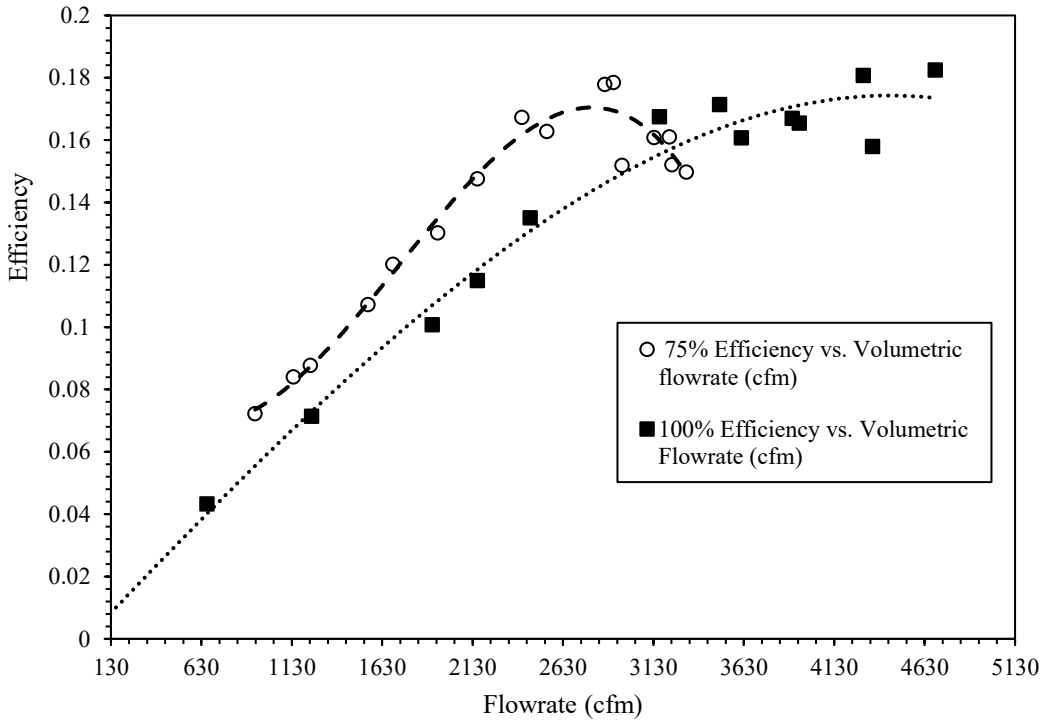


Figure 4. Efficiency vs Flowrate fit with a third-order polynomial curve

Using Equations 8 and 9, the dimensionless head rise and flowrate coefficients were calculated and are plotted against each other for each rotational speed in the graph below.

The measured 1320 rpm performance data were scaled to 1750 rpm using turbomachinery similarity relationships ($Q \propto N$, $H \propto N^2$, $P \propto N^3$), assuming constant geometry and dynamically similar flow. A detailed example of the scaling procedure is included in Attachment C.

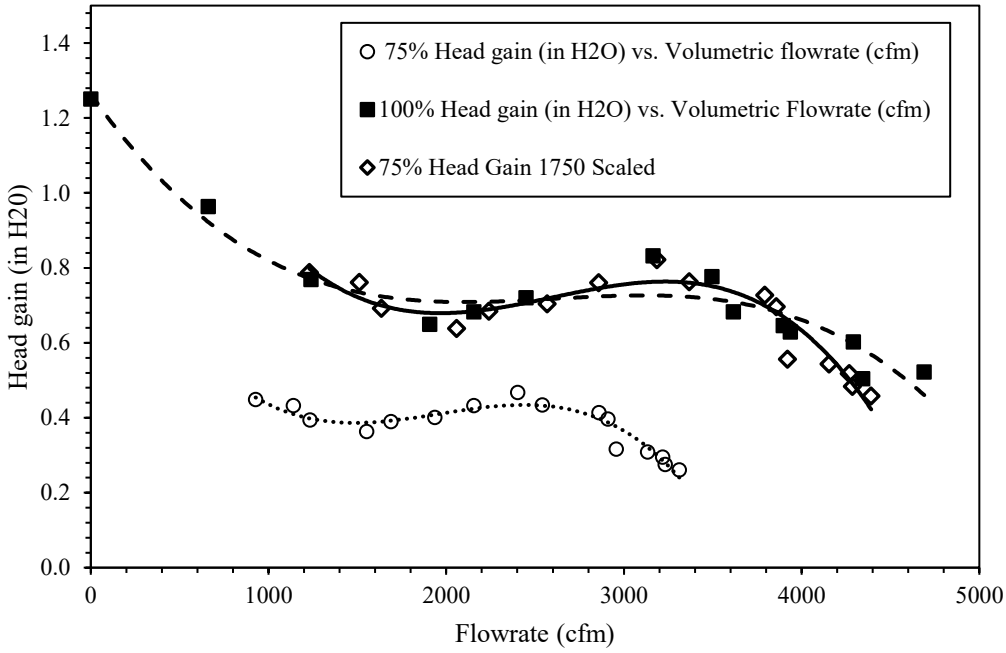


Figure 5. Head Gain Vs. Flow Rate with Overlayed Scaled/Predicted 1750 rpm Head Gain

Figure 5 compares the measured head gain at 75% (1320 rpm) and 100%(1750(rpm) with the predicted 1750 rpm head obtained using the fan similarity laws. The scaling relationship: $H \propto N^2$ was applied to the 1320 rpm, data using a speed ratio or r- 1750/1320=1.326. The predicted head values closely follow the measured 1750 rpm data over the mid-range flowrates. Minor deviations are observed at very low and very high flowrates, where stall and increased losses reduce dynamic similarity. Overall, the agreement between predicted and measured head supports the validity of the quadratic speed scaling relationship.

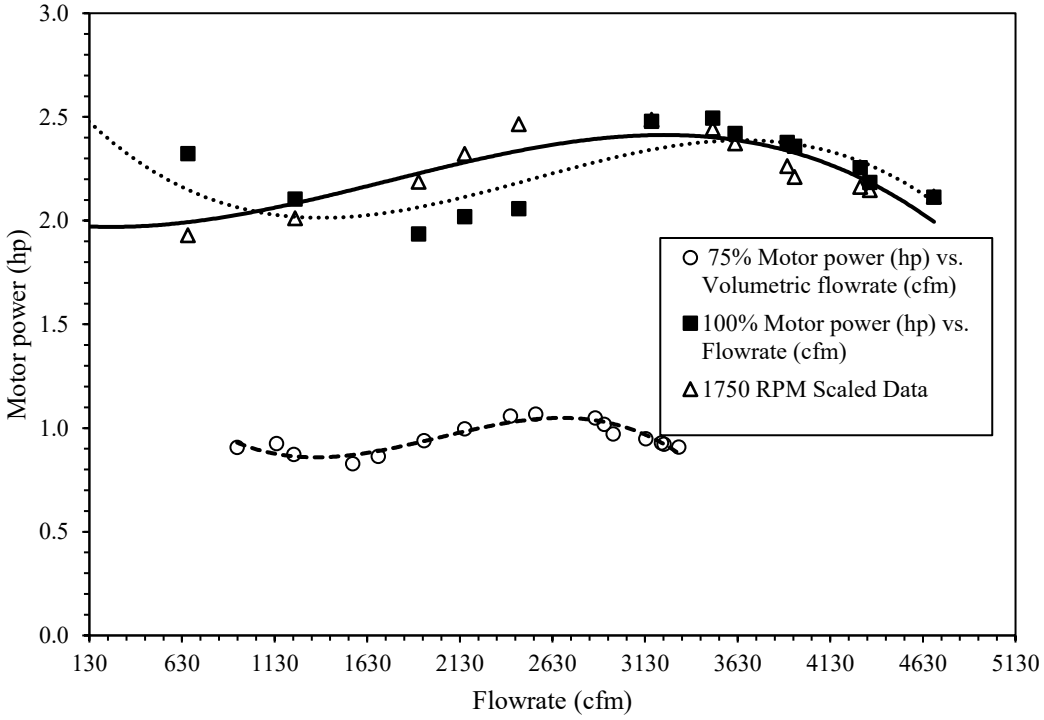


Figure 6. Motor Power Vs. Flow Rate with Overlayed Scaled/Predicted 1750 rpm Motor Power

Figure 6 shows the measured motor power at both operating speeds along with the predicted 1750 rpm power obtained using the cubic scaling relationship $P \propto N^3$. The predicted power values generally follow the trend of the measured 1750 rpm data, particularly in the mid-range of flowrates. Some deviation is observed at low flow conditions, likely due to non-ideal flow behavior near stall and electrical losses in the motor. The overall trend confirms the expected cubic dependence of power on rotational speed.

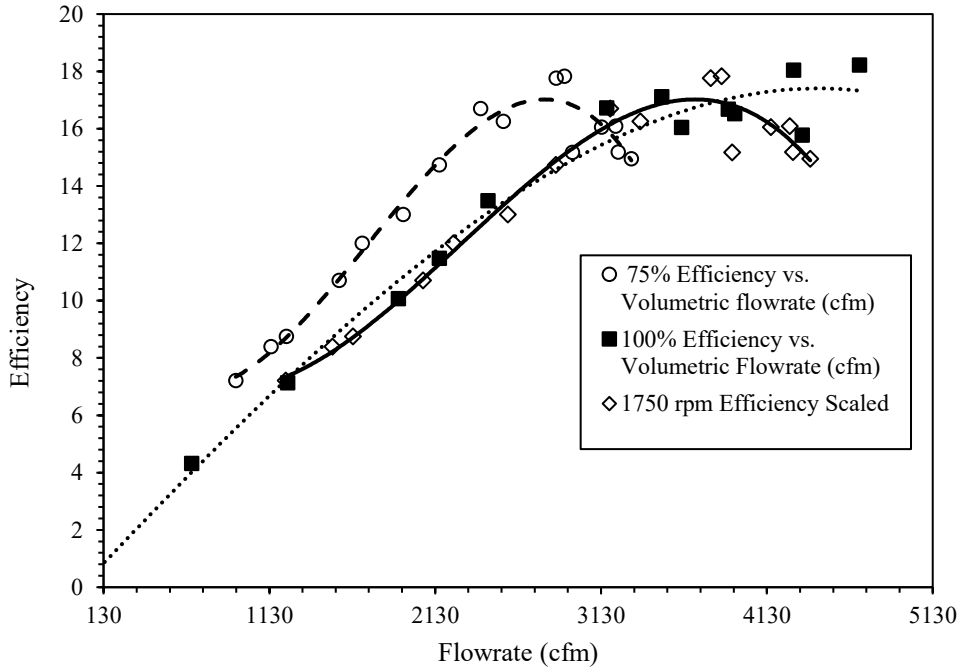


Figure 7. Motor Power Vs. Efficiency with Overlayed Scaled/Predicted 1750 rpm Efficiency

Figure 7 compares the measured efficiency curves at 75% and 100% speed with the predicted 1750 rpm efficiency obtained from similarity laws. Since efficiency is dimensionless, the fan laws predict that efficiency should remain approximately constant with changes in rotational speed. The predicted efficiency curve shows reasonable agreement with the measured 1750 rpm data in the mid-range flow region. Differences near stall and high flow conditions are expected due to increased losses and departure from ideal dynamic similarity. The results indicate that efficiency remains approximately independent of speed over the normal operating range, consistent with turbomachinery similarity theory.

The best agreement between predicted and measured data occurs near the best efficiency point (BEP), where the flow is most stable and losses are minimized. Larger deviations near stall indicate that similarity assumptions break down under highly separated or unsteady flow conditions.

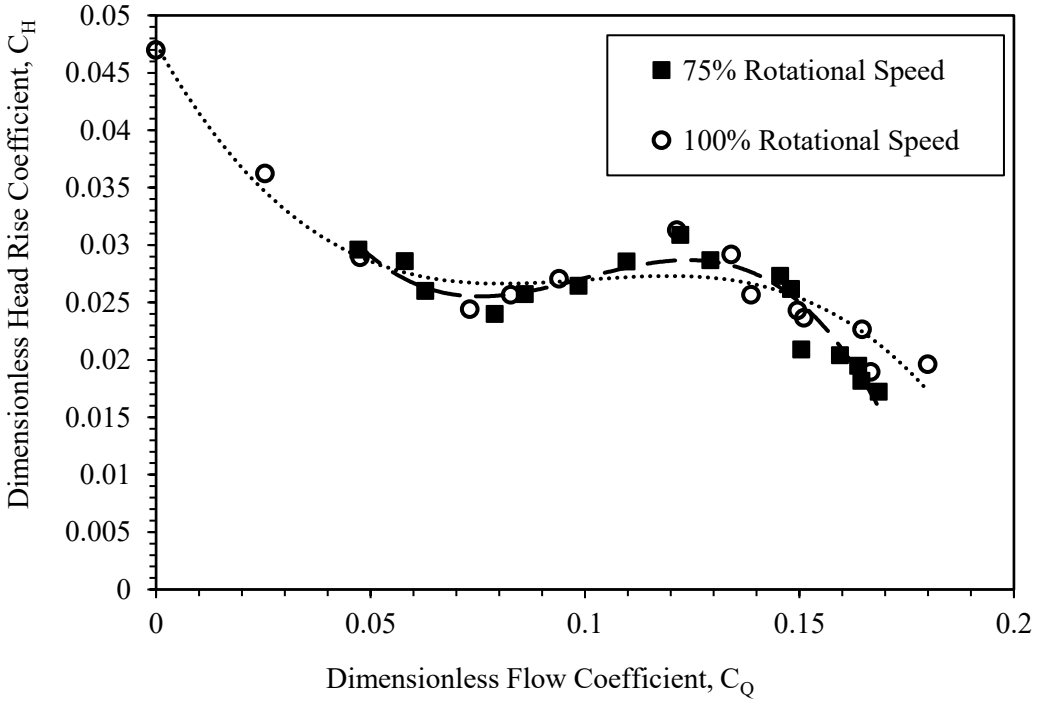


Figure 8. Dimensionless numbers at each rotational speed fit with a third-order polynomial curve.

When a trendline is added to the data at each rotational speed, the two datasets collapse into essentially a single curve, which is expected due to turbomachinery similarity. Because the two datasets rely on the same fluid and the same machine (corresponding to the same diameter of fan) and vary only with rotational speed, the performance much follow the affinity laws for turbomachinery, where $Q \propto \omega$ and $h_a \propto \omega^2$. When the speed dependence is removed by nondimensionalizing values, the curve becomes machine-specific and independent of speed, which is shown by the plot in Figure 7 (essentially, $C_H = f(C_Q)$).

To demonstrate that our incompressible flow assumption is valid, we can employ the Mach number. If the Mach number is less than 0.3, we know that the density changes in the fluid are less than 5%, so our incompressible flow assumption is validated. Plugging in our average velocity ($8\,ft/s$) and the speed of sound in air ($1125\,ft/s$), we get:

$$Ma = \frac{V}{c} = \frac{8\frac{ft}{s}}{1125\frac{ft}{s}} = 0.0071 < 0.3,$$

and therefore, the incompressible assumption is valid.

References

[1] ME 347 Lab 10 Manual, Arturo Cajal.

List of Attachments

A. Table of Experimental Data
B. Sample Calculations: Volumetric Flowrate, Head Gain, Motor Power, Efficiency
C. Sample Calculations: Scaled Volumetric Flowrate, Head Gain, Motor Power, Efficiency

ENGINEERING DATA SHEET	Sheet No.	A – 1/1	Date:
Attachment A: Table of Experimental Data	Prepared	KG	02/16/26
	Checked	GL	02/16/26
	Approved	LMS	02/16/26

Date: 02/09/2026 RPM: 1320 RPM, 1750 RPM Pamb: 14.627 psi Tamb: 70.0 °F				Engineers: Kelly Graham, Garrett Leighton, Gabriel Choi, Laura Strobbe		
No.	Static Pressure (in H20)	Motor Voltage (Volts-AC)	Motor Current (Amps)	Left Flow (in H20)	Right Flow (in H20)	Comments
75% (1320 RPM)						
1	0.3	157	2.45	0.13	0.12	
2	0.35	158	2.47	0.14	0.1	
3	0.4	158.5	2.48	0.15	0.09	
4	0.45	160.7	2.5	0.18	0.06	
5	0.5	162.5	2.53	0.21	0.03	
6	0.7	165.8	2.6	0.23	0.02	
7	0.75	168.8	2.63	0.26	0.01	
8	0.85	169.9	2.66	0.29	0	
9	0.95	169.7	2.64	0.26	0	
10	0.9	166.7	2.53	0.21	0	
11	0.85	159	2.5	0.17	0	
12	0.85	151	2.42	0.13	0	
13	0.8	150.5	2.33	0.11	0	
14	0.9	154	2.4	0.07	0	
15	1	159.9	2.45	0.06	0	
16	1.05	157.3	2.44	0.04	0	
100% (1750 RPM)						
1	0.6	274.5	3.26	0.235	0.235	
2	0.65	277.7	3.33	0.245	0.185	
3	0.9	280.9	3.4	0.39	0.085	
4	1.05	289.5	3.45	0.41	0.04	
5	1.1	290	3.47	0.415	0.035	
6	1.25	292.8	3.5	0.45	0.01	
7	1.5	297.5	3.55	0.55	0	
8	1.7	297.3	3.53	0.45	0	
9	1.55	266.5	3.27	0.27	0	
10	1.5	274.8	3.11	0.21	0	
11	1.45	262.8	3.12	0.165	0	
12	1.8	274	3.25	0.07	0	
13	2.3	285.1	3.45	0.02	0	
14	3	297.7	3.55	0	0	

Table 3. Experimental Data for 1320 and 1750 RPM

ENGINEERING DATA SHEET	Sheet No.	B – 1/1	Date:
Attachment B: Sample Calculations: Volumetric Flowrate, Head Gain, Motor Power, Efficiency	Prepared	KG	02/16/26
	Checked	LMS	02/16/26
	Approved	GC	02/16/26

The following is an example calculation for the volumetric flowrate, head gain, motor power, and efficiency for number 1 when the fan speed is set to 1320 RPM.

The measured head (in inches of water) in the left and right duct is converted into feet of air.

$$h_{a,air} = \left(\frac{\rho_{H_2O}}{\rho_{air}}\right) h_{a,H_2O}$$

$$h_8 = \left(\frac{62.4 lbm/ft^3}{0.0746 lbm/ft^3}\right) 0.12\ in \left(\frac{1\ ft}{12\ in}\right) = 8.37\ ft$$

$$h_9 = \left(\frac{62.4 lbm/ft^3}{0.0746 lbm/ft^3}\right) 0.13\ in \left(\frac{1\ ft}{12\ in}\right) = 9.07\ ft$$

The center-line velocity was calculated using the relationship between dynamic pressure and the velocity.

$$V_C = \sqrt{\frac{2(p_T - p)}{\rho_{air}}}$$

$$(p_T - p) = \rho_{H_2O} g h_{H_2O}$$

$$(p_T - p)_8 = \left(\frac{62.4 lbm}{ft^3}\right)(32.174 \frac{ft}{s^2})(0.12 in)\left(\frac{1\ ft}{12\ in}\right)$$

$$(p_T - p)_9 = \left(\frac{62.4 lbm}{ft^3}\right)(32.174 \frac{ft}{s^2})(0.13\ in)\left(\frac{1\ ft}{12\ in}\right)$$

$$V_{C_8} = \sqrt{\frac{2\left(\frac{62.4 lbm}{ft^3}\right)(32.174 \frac{ft}{s^2})(0.12 in)\left(\frac{1\ ft}{12\ in}\right)}{0.0746\ lbm/ft^3}} = 23.2\ ft/s$$

$$V_{C_9} = \sqrt{\frac{2\left(\frac{62.4 lbm}{ft^3}\right)(32.174 \frac{ft}{s^2})(0.13 in)\left(\frac{1\ ft}{12\ in}\right)}{0.0746\ lbm/ft^3}} = 24.1\ ft/s$$

The average velocity was calculated using the relationship between the average and centerline velocity.

$$\frac{\bar{V}}{V_c} = \frac{2n^2}{(n+1)(2n+1)}, \quad n = -1.7 + 1.8\log\left(Re_{V_c}\right)$$

$$Re_{V_c} = \frac{\rho V_c D_{duct}}{\mu}$$

$$Re_{V_{c_8}} = \frac{(0.0746 \frac{lbm}{ft^3})(23.2 \frac{ft}{s})(16.0\ in)}{3.37\ x\ 10^{-7} lbfs/ft^2} = 212592$$

$$Re_{V_{c9}} = \frac{(0.0746\frac{lbm}{ft^3})(24.1\frac{ft}{s})(16.0\ in)}{3.37\ x\ 10^{-7} lbfs/ft^2} = 221273$$

$$n_8 = -1.7 + 1.8\log(212592) = 7.89$$

$$n_9 = -1.7 + 1.8\log(221273) = 7.92$$

$$\bar{V}_8 = \frac{2(7.89)^2}{(7.89+1)(2(7.89)+1)}\left(23.2\frac{ft}{s}\right) = 19.4\ ft/s$$
$$\bar{V}_9 = \frac{2(7.92)^2}{(7.92+1)(2(7.92)+1)}\left(24.1\frac{ft}{s}\right) = 20.2\ ft/s$$

The total velocity, which is the sum of the average velocity through the left and right duct, was use to calculate the volumetric flowrate along with the area of the duct.

$$Q_{fan} = (\bar{V}_8 + \bar{V}_9)A_{duct}$$
$$Q_{fan} = (19.4ft/s + 20.2ft/s)1.396ft^2\left(\frac{60\ s}{1\ min}\right) = 3312\ cfm$$

The head was calculated using the following equation, assuming constant elevation ($\Delta z = 0$), ambient atmospheric air at the inlet ($p_1 = p_{atm}, \bar{V}_1 \approx 0$), and low pressure and high velocity air at the outlet ($\alpha_2 \approx 1$). The static pressure and the ambient pressure in the room were converted to feet of air.

$$h_a = \frac{\Delta p_{fan}}{\rho g} + (1 + K_L)\left(\frac{\bar{V}_{fan}{}^2}{2g}\right)$$
$$h_{a,ft\ air} = \frac{28283.7ft - 28262.8ft}{0.0746\frac{lbm}{ft^3}(32.174\frac{ft}{s^2})} + (1+0.02)\left(\frac{(22.52\frac{ft}{s})^2}{2(32.174\ ft/s^2}\right)$$
$$h_{a,\,in\,H_2O} = \left(\frac{\rho_{air}}{\rho_{H_2O}}\right)h_{a,air}$$
$$h_{a,\,in\,H_2O} = \left(\frac{0.0746\frac{lbm}{ft^3}}{62.4\ \frac{lbm}{ft^3}}\right)18.18ft = 0.261\ in\ H_2O$$

Next, the motor power was calculated.

$$\dot{W}_{shaft} = \sqrt{3}V_{LL}I_L\cos\theta$$

$$\dot{W}_{shaft} = \sqrt{3}(157\ V)(2.45\ A)(0.75) = 499.675\frac{1\ hp}{550\ lbfft/s} = 0.91\ hp$$

The efficiency was calculated by determining the ratio between the power delivered to the fluid and the motor power.

$$\eta = \frac{\mathcal{P}_f}{\dot{W}_{shaft}} = \frac{\rho g Q_{fan} h_a}{\sqrt{3}V_{LL}I_L\cos\theta}$$

$$\eta = \frac{\left(0.00232\,\frac{slug}{ft^3}\right)\left(32.174\,\frac{ft}{s^2}\right)\left(\frac{3312cfm}{60s}\right)18.18\,ft}{499.68\,lbfft/s} = 15.0\,\%$$

ENGINEERING DATA SHEET	Sheet No.	C – 1/1	Date: 02/16/26
Attachment C: The following is an example calculation showing how the 1320 rpm data point is scaled to predict performance at 1750 rpm using the fan similarity laws.	Prepared	GC	02/16/26
	Checked	LMS	02/16/26
	Approved	KG	02/16/26

Given (from 1320 rpm, Run #1):

- N1 = 1320 rpm
- N2 = 1750 rpm
- Q1 = 3312 cfm
- H1 = 0.81117 in H2O
- P1 = 475.196 W
- η1 = 0.2502 (25.02%)

1. Speed Ratio

$$r = \frac{N_2}{N_1}$$
$$r = \frac{1750}{1320} = 1.32576$$
$$r^2 = (1.32576)^2 = 1.75764$$
$$r^3 = (1.32576)^3 = 2.330\ (rounded)$$

2. Scale Flowrate (Q proportional to N)

$$Q_2 = Q_1 * r$$
$$Q_2 = (3312\ cfm)(1.32576)$$
$$Q_2 = 4391\ cfm\ (predicted\ at\ 1750\ rpm$$

3. Scale Head Gain (H proportional to N^2)

$$H_2 = H_1 * r\text{^}2$$
$$H_2 = (0.81117\ in\ H2O)(1.75764)$$
$$H_2 = 1.4257\ in\ H2O\ (predicted\ at\ 1750\ rpm)$$

4. Scale Power (P proportional to N^3)

$$P_2 = P_1 * r\text{^}3$$
$$P_2 = (475.196\ W)(2.330)$$
$$P_2 = 1107.3\ W\ (predicted\ at\ 1750\ rpm)$$

5. Scale Efficiency (Efficiency approximately constant)

$$\eta_2 \approx \eta_1$$
$$\eta_2 \approx 0.2502 = 25.02\%$$

Predicted 1750 rpm Results (Scaled from 1320 rpm Run #1)

- $Q_2 = 4391\ cfm$
- $H_2 = 1.4257\ in\ H2O$
- $P_2 = 1107.3\ W$
- $\eta_2 \approx 25.02\%$

These predictions are based on fan similarity laws assuming the same fan geometry and dynamically similar flow. Deviations are expected near stall/low-flow conditions where separation and losses increase.